

Diffusion Policy: Visuomotor Policy Learning via Action Diffusion

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<https://diffusion-policy.cs.columbia.edu>

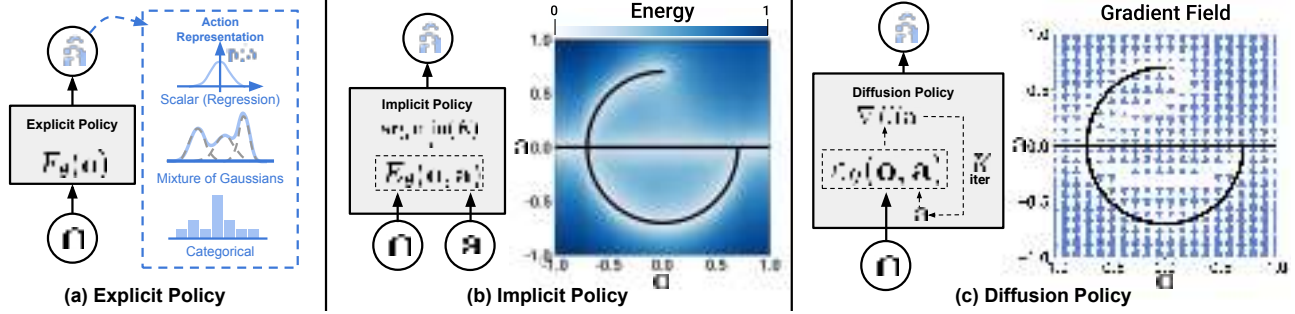


Fig. 1: **Policy Representations.** a) Explicit policy with different types of action representations. b) Implicit policy learns an energy function conditioned on both action and observation and optimizes for actions that minimize the energy landscape c) Diffusion policy refines noise into actions via a learned gradient field. This formulation provides stable training, allows the learned policy to accurately model multimodal action distributions, and accommodates high-dimensional action sequences.

Abstract—This paper introduces Diffusion Policy, a new way of generating robot behavior by representing a robot’s visuomotor policy as a conditional denoising diffusion process. We benchmark Diffusion Policy across 12 different tasks from 4 different robot manipulation benchmarks and find that it consistently outperforms existing state-of-the-art robot learning methods with an average improvement of 46.9%. Diffusion Policy learns the gradient of the action-distribution score function and iteratively optimizes with respect to this gradient field during inference via a series of stochastic Langevin dynamics steps. We find that the diffusion formulation yields powerful advantages when used for robot policies, including gracefully handling multimodal action distributions, being suitable for high-dimensional action spaces, and exhibiting impressive training stability. To fully unlock the potential of diffusion models for visuomotor policy learning on physical robots, this paper presents a set of key technical contributions including the incorporation of receding horizon control, visual conditioning, and the time-series diffusion transformer. We hope this work will help motivate a new generation of policy learning techniques that are able to leverage the powerful generative modeling capabilities of diffusion models. Code, data, and training details will be publicly available.

I. INTRODUCTION

Policy learning from demonstration, in its simplest form, can be formulated as the supervised regression task of learning to map observations to actions. In practice however, the unique nature of predicting robot actions — such as the existence of multimodal distributions, sequential correlation, and the requirement of high precision — makes this task distinct and challenging compared to other supervised learning problems.

Prior work attempts to address this challenge by exploring different *action representations* (Fig 1 a) – using mixtures of Gaussians [29], categorical representations of quantized actions [42], or by switching the *the policy representation* (Fig

1 b) – from explicit to implicit to better capture multi-modal distributions [12, 56].

In this work, we seek to address this challenge by introducing a new form of robot visuomotor policy that generates behavior via a “conditional denoising diffusion process [18] on robot action space”, **Diffusion Policy**. In this formulation, instead of directly outputting an action, the policy infers the action-score gradient, conditioned on visual observations, for K denoising iterations (Fig. 1 c). This formulation allows robot policies to inherit several key properties from diffusion models – significantly improving performance.

- **Expressing multimodal action distributions.** By learning the gradient of the action score function [46] and performing Stochastic Langevin Dynamics sampling on this gradient field, Diffusion policy can express arbitrary normalizable distributions [32], which includes multimodal action distributions, a well-known challenge for policy learning.
- **High-dimensional output space.** As demonstrated by their impressive image generation results, diffusion models have shown excellent scalability to high-dimension output spaces. This property allows the policy to jointly infer a *sequence* of future actions instead of *single-step* actions, which is critical for encouraging temporal action consistency and avoiding myopic planning.
- **Stable training.** Training energy-based policies often requires negative sampling to estimate an intractable normalization constant, which is known to cause training instability [11, 12]. Diffusion Policy bypasses this requirement by learning the gradient of the energy function and thereby achieves stable training while maintaining